

RETINAL EDGE TRIAGE

Explainable AI-Assisted Diabetic Retinopathy Screening for Rural Healthcare
Official Clinical Triage Report | Generated: 04-Sep-2026 16:37 UTC

Patient ID: RET-2026-516077	Full Name: Keshav	Age / Gender: 88 / Female
Screening ID: SCR-2026-000036	Location: PHC Rampur, Rural District	Diabetes Type / Duration: Type 2 (5 years)

1. Retinal Image Acquisition & Quality Gate

Eye Evaluated: Right Eye	Image Source: Fundus Camera	Quality Score: 56.0%	Quality Status: BORDERLINE
Focus / Blur: 17.4%	Illumination: 31.8%	Vessel Visibility: 53.3%	Field of View: 92.2%

2. AI Screening Assessment (APTOS 2019 & IDRiD Models)

DR Severity Grade:	Grade 2 — Moderate DR	Referable DR:	YES (Action Required)
AI Confidence:	96.6%	DME Risk Status:	Possible DME detected (Mild exudative clustering noted near temporal arcade. Clinical macular assessment recommended.)
Detected Lesions (IDRiD):	Microaneurysms: 4 Hemorrhages: 2 Hard Exudates: 2 Soft Exudates: 1	Model Version:	Enhanced Clinical Ensemble v2.0 (Lesion-Grounded Fusion + TTA)

3. Human-in-the-Loop Ophthalmologist Adjudication

Doctor Decision:	ACCEPTED	Reviewing Doctor:	Dr. Ophthalmologist
Final Grade Assigned:	Grade 2	Override Justification:	None
Clinical Notes:	Clinical findings are consistent with non-proliferative diabetic retinopathy.	HCW Instructions:	None

4. Risk Triage & Clinical Referral Plan

Risk Category:	HIGH RISK	Referral Priority:	URGENT
Referral ID:	REF-2026-00017	Recommended Facility:	District Eye Hospital, Bareilly
Triage Reason:	High Risk: Grade 2 with DME suspicion.	Follow-up Schedule:	1 Months

STATUTORY CLINICAL NOTICE: This report is generated by an artificial intelligence screening triage prototype (Retinal Edge Triage). AI predictions DO NOT constitute a definitive medical diagnosis. Final clinical interpretation and management must always be performed by a qualified ophthalmologist. Compliant with rural screening clinical governance principles.